

Are All Instances Equally Explainable? A Study on the Impact of Locality on the Explainability of Automatic Text Classification Tasks

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Abstract. Many machine learning (ML) models still operate as “black boxes”, limiting users’ ability to understand and trust automated decisions. While Explainable Artificial Intelligence (XAI) has advanced, most approaches emphasize model mechanisms or explanation strategies, paying little attention to how the *structure of the data* itself shapes explanation quality. In this work, we explore the role of *locality* in shaping the perceived explanation quality for Automatic Text Classification (ATC), where locality consists of how instances are positioned relative to neighbors in representation space. We propose a *data-centric framework* that characterizes locality using two structural factors: (i) the average similarity between an instance and its neighbors and (ii) the entropy of the local class distribution. Based on these factors, we generate explanations grounded in neighborhood structure through locality-based word clouds and compare them to attention-based explanations. Unlike prior work that typically assumes locality inherently improves explainability, our framework systematically investigates explainability as a function of the *spatial organization of data representations*, while also establishing a principled way to relate local data characteristics to human-perceived explanatory quality. A user study with over 40 participants across two datasets evaluates explanations under criteria of *Satisfaction*, *Completeness*, and *Trust*. Results reveal that instances in highly homogeneous regions (high similarity, low entropy) produce explanations that users consistently rate as clearer and more useful, with statistically significant differences across datasets. These findings highlight the importance of locality as a key factor in explainability, offering insights for the design of explanation strategies in ATC and broader XAI research.

Keywords: Explainable Artificial Intelligence · Instance Locality · Human-grounded Evaluation · Automatic Text Classification.

1 Introduction

Despite high predictive performance and omnipresence in linguistic applications [12], many ML models are considered “black boxes” due to their complex, opaque processes [35]. Opacity can hinder user trust, bias detection, and error analysis, and has ethical implications [23]. Regulations such as the General Data Protection Regulation (GDPR) and the EU Artificial Intelligence Act reinforce demands for transparency in automated decision-making [17, 21]. Explainability fosters trust and supports model debugging and performance refinement [12, 63].

Extensive efforts have addressed model explainability either through: (i) post-hoc explanations, which interpret model outputs without internal access; or intrinsically interpretable models, designed for transparency by construction [1, 39]. A recurring assumption across post-hoc techniques, including LIME and its extensions as well as SHAP and related methods [18, 19, 41, 52, 57], is that *locality*, the set of similar data instances in a feature space, facilitates explainability [12, 40, 47, 68]. While widely adopted, this assumption is typically treated as self-evident: locality is presumed beneficial, yet rarely examined as an explicit object of analysis, nor linked to how explanations are actually perceived by users.

As a consequence, the role of locality has largely remained untested through systematic, user-centered empirical analysis. In particular, it is still unclear whether and under which conditions, the structural properties of local neighborhoods translate into consistent differences in how users perceive explanation quality. Rather than assuming locality as inherently beneficial, this work grounds explainability in measurable geometric characteristics of the embedding space, aiming to clarify when and how locality constitutes a meaningful and reliable factor in supporting user understanding and evaluation of explanations.

Building on this motivation, we tackle this literature gap by explicitly investigating the role of locality on the explainability of Automatic Text Classification (ATC) tasks. This paper investigates the hypothesis that *more homogeneous localities, with higher similarity and lower entropy, yield more interpretable outcomes*. We characterize an instance’s locality by two structural factors: the *average similarity* between an instance and its neighbors and the *entropy* of the local class distribution. In order to investigate the hypothesis, we raise the following research questions, seeking to empirically answer them: (RQ1) *How can locality be analyzed and characterized from a perspective of ML explainability?* (RQ2) *Do locality-based explanations contribute to the explainability of the model’s results?*

Towards answering the RQs, we (i) compute neighborhoods using transformer-based embeddings [33] and a k -nearest neighbors (kNN) algorithm; (ii) characterize instance locality based on similarity and entropy, and employ a heatmap to visualize the distribution of instances according to this characterization; and (iii) select representative instances with high and low homogeneity, guided by regions identified in the heatmap, and generate corresponding word cloud explanations, aiming to analyze the impact of locality on them. Furthermore, we generated attention word cloud explanations for each instance to compare locality and attention explanations [64] using the same visual metaphor.

To evaluate the generated explanations, we conduct a user study with over 40 participants on two different datasets: AGNews, focused on news and ACM, restricted to Computer Science research articles. The study protocol was approved by the University’s research ethics committee. Our study tackle three evaluation criteria [28]: *Satisfaction* (user’s judgment that an explanation is clear, useful, and fits their goals.), *Completeness* (how fully the explanation covers key factors and decision details), and *Trust* (confidence to know when to rely on the system and when to be cautious, shaped by explanations). Results show that explanations for instances in *highly-homogeneous localities* received consistently higher ratings than those from low-homogeneous ones, regardless of classification correctness, with statistically significant differences across datasets. When comparing explanation strategies, *locality word clouds* performed comparably to *attention-based* ones in AGNews but were significantly preferred in ACM, highlighting the particular value of locality in more specialized or complex domains.

The explanations generated by the proposed locality-based methodology are intended for machine learning practitioners and researchers, i.e., users who engage with ML models at some technical level seeking to understand model behavior for tasks such as debugging, error analysis, comparative assessment of explanation mechanisms, and reflective understanding of model decisions.

In summary, our contributions are threefold:

1. We formulate and investigate a novel research question by examining whether text instance classifications exhibit varying degrees of explainability and how these variations are influenced by the homogeneity of their local neighborhoods, thereby addressing a gap that remains largely unexplored in current Explainable Artificial Intelligence(XAI) research.
2. We propose a data-centric methodological framework to characterize the locality of instances through entropy and average similarity, generating explanations based on neighborhood structure. Thus, our approach is among the few in the literature that systematically investigates explainability as a function of the spatial structure of data representations, rather than treating it solely as a property of models or explanatory techniques. Moreover, the framework provides a systematic approach for relating local data properties to human-perceived explanation quality.
3. We conduct a comprehensive empirical user study assessing explanation quality across three user-centric dimensions: *Satisfaction*, *Completeness*, and *Trust*. By contrasting a locality-based explanation approach with an attention-based baseline, we provide empirical evidence and novel insights into the conditions under which locality improves text classification explainability, thereby complementing and extending current understanding of XAI in ATC.

2 Background and Related Work

2.1 Explainability and Interpretability

The widespread adoption of ML across NLP tasks has intensified the demand for models that are not only accurate but also understandable, motivating substantial advances in XAI aimed at transparency, accountability, and responsible

deployment [12, 63, 65]. In the literature, *explainability* and *interpretability* are often used interchangeably to denote the goal of rendering model behavior understandable through human-interpretable representations [46, 68]. Other work draws a sharper distinction, defining interpretability as an intrinsic property of models whose mechanisms are directly comprehensible, and explainability as the generation of post-hoc explanations for opaque models [2, 42]. Following this distinction, the present work is situated within explainability, as it seeks to support understanding of model decisions without requiring inherently interpretable classifiers. In this context, *faithfulness* captures the degree to which explanations accurately reflect the model’s internal reasoning [31, 38].

An *explanation* describes how a system arrives at a decision by exposing aspects of its reasoning or mechanisms, whereas a *justification* provides a normative account of why a decision is appropriate with respect to deployment goals and values [36]. Despite extensive progress, designing effective explanations remains challenging due to user heterogeneity, presentation choices, and the inherent trade-off between transparency and predictive performance [14, 23, 58].

We position this work within the explainability landscape by examining how the *local structure of data* shapes the perceived quality of explanations. Rather than probing internal model signals, we adopt a user-centered, data-driven strategy that leverages locality-based word clouds to convey how an instance’s locality contributes to its classification and explanation. While this improves explainability and perceived clarity, it may involve a trade-off with faithfulness, as explanations are not directly tied to the model’s internal computations [31, 38].

2.2 Locality

Locality refers to proximity relationships in a feature space, where similar instances occupy nearby regions. In this work, locality is defined as a property of the *empirical data distribution*, rather than as an artifact of synthetic perturbations. This notion underlies classical methods such as k -nearest neighbors and k -means, and aligns with van Rijsbergen’s clustering hypothesis in Information Retrieval [54]. Under this view, neighborhoods reflect the geometric and semantic structure of the observed data manifold.

Locality also plays a central role in XAI, though most methods do not distinguish between neighborhoods drawn from the data distribution and those induced by model-centric perturbations. Consequently, locality is often assumed to improve interpretability without systematic analysis of when this assumption holds or how locality is operationalized.

Our work addresses this gap by focusing explicitly on *data distributional locality* and by studying how empirical neighborhood structure influences the perceived comprehensibility of explanations for automatic text classification.

Post-hoc explanation methods exploiting locality can be broadly grouped into three families. **Local surrogate models**, such as LIME and its variants [18, 19, 51–53, 57], approximate decision boundaries using interpretable models fitted on locally sampled regions, typically constructed via synthetic perturbations. **Perturbation-based methods** estimate feature influence by modifying

inputs and observing prediction changes, implicitly assuming that small perturbations preserve meaningful neighborhoods [4, 13, 16, 20, 47, 56]. **Counterfactual explanations** identify minimal changes that flip predictions [13, 22, 47], while **game-theoretic attribution methods**, such as SHAP [19, 41], rely on local reference populations to stabilize attributions.

Despite methodological differences, these approaches share an implicit assumption: instances deemed local, whether real or synthetic, should exhibit homogeneous model behavior. However, the literature offers limited analysis of when locality grounded in the empirical data distribution, rather than neighborhoods induced by perturbation, actually improves explainability or faithfulness.

In contrast, we operationalize locality strictly over observed data by characterizing neighborhoods via average similarity and class-distribution entropy. We empirically examine how variations in neighborhood homogeneity affect the *perceived quality and comprehensibility* of explanations. This data-centric perspective clarifies the distinction between model-centric and data distributional locality and grounds our analysis in principles from statistical learning theory. Notably, our setting also differs from training-dynamics analyses (e.g., Dataset Cartography [62]), as explainability arises from semantic position in the embedding space at inference time.

2.3 Entropy

Entropy quantifies uncertainty in a system [59]. In feature spaces, neighborhoods with highly similar instances tend to exhibit lower entropy, whereas heterogeneous neighborhoods display higher uncertainty. Prior work has explored the relationship between entropy, neighborhood structure, and feature relevance [5, 8, 30, 43], including neighborhood-based entropy measures [30].

In this work, entropy captures the class-distribution uncertainty within local neighborhoods. Together with average similarity, it characterizes locality structure and enables analysis of how local homogeneity influences the perceived clarity and usefulness of explanations.

In summary, our contribution departs from prior XAI work by adopting a data-centric view of explainability. Rather than assuming that locality inherently improves explanations, we explicitly characterize neighborhood structure and study its impact on user perception. We introduce a locality-based explanatory framework, compare it with attention-based explanations under a unified visual metaphor, and provide user-study evidence across two datasets linking local homogeneity to clearer and more useful explanations, thereby positioning locality as a key explanatory factor in XAI.

3 Study Methodology

This section describes the complete methodology of the study, including the instantiation of a general locality-based framework for analyzing explainability and generating explanations based on neighborhood structure, the experimental

setup for the user study design, and the user-centered evaluation protocol ³. Before detailing the methodology, we first describe in the following subsection the datasets used to instantiate and evaluate the proposed framework.

3.1 Experimental Datasets

The datasets here described are used to instantiate and evaluate the proposed framework and do not restrict its generality. Dataset selection was guided by both methodological and cognitive constraints. Specifically, suitable datasets must present:

1. A moderate number of classes to simultaneously induce meaningful variation in neighborhood entropy and avoid overburdening participants with the analysis of an excessively large class set.
2. Texts with medium length (on average between 200 and 450 characters) to prevent cognitive overload during explanation inspection in the user study.
3. Distinct domains coverage, enabling the examination of locality effects across different topical and semantic structures.

Together, these criteria substantially restrict the pool of applicable benchmarks, making dataset selection a critical component of the experimental design and an enabling factor for the proposed locality-based explainability analysis. To assess the methodology and its generalizability, we conducted experiments on two text classification datasets: AGNews and ACM.

AGNews: A widely used benchmark for topic classification comprising news articles aggregated from more than 2,000 sources. In the subset used in our experiments, each article contains a title and a short description and is assigned to one of four classes: *World*, *Sports*, *Business*, or *Science/Technology* [24] (for the complete class definitions, see Appendix A).

ACM: A curated subset of the ACM Digital Library comprising research articles in Computer Science. To ensure a feasible user study and maintain evaluation comparability with AGNews, we selected four representative categories from the eleven available in the original dataset: *Hardware* (Hw), *Software* (Sw), *Theory of Computation* (Tc), and *Information Systems* (Is) (complete definitions in Appendix A). Each instance includes both the title and abstract. These four classes were chosen for their semantic diversity and adequate representation in the dataset (*Hw*: 8%, *Sw*: 26%, *Tc*: 4%, *Is*: 19%), allowing representative sampling while mitigating potential bias from overly similar categories.

By selecting datasets with diverse vocabularies, styles, and domains, we provide initial evidence of robustness of our approach, strengthening the empirical basis and indicating its applicability beyond a single dataset. The two datasets represent distinct domains, both comprise a reasonable number of classes and contain news articles and scientific texts of moderate length, thereby satisfying the defined constraints.

³ Code and experimental materials available:
<https://github.com/lizianessoares/locality-explainability>

3.2 Framework for local-based explanations

Our framework include a sequence of steps to represent and evaluate the locality of text instances. Each document is encoded using a transformer-based model (BERT) [33] to obtain contextual embeddings, from which we apply k -nearest neighbors (k NN) to estimate locality. For each instance, we compute the Shannon entropy [59] of its neighborhood’s class distribution and the average similarity to its neighbors. A heatmap of these values enables the identification of regions with *high* and *low locality homogeneity*, from which we sample diverse instances.

For each instance, we generate two explanations: a locality-based explanation and an attention-based explanation. In both cases, explanations are presented as word clouds, where word size reflects frequency (in locality) or attention weight, supporting rapid recognition of salient terms. Colors are used only for visual differentiation and do not encode additional information. We evaluate explanations in an online user study, assessing perceived quality across key dimensions. Locality characterization addresses RQ1, while instance sampling, explanation generation, and the user study address RQ2.

Locality Characterization

We use Transformer-based contextual document representations generated by BERT [33], which enhance class separability and support more homogeneous local neighborhoods, potentially improving the quality of local explanations. Prior work [3] has shown that such representations create highly separable spaces for ATC, with minimal differences compared to other Transformer-based encoders such as RoBERTa and XLNet [67]. We selected BERT over RoBERTa, a very effective Small Language Model (SLM) [11] comparable to most Large LMs, because its *WordPiece* tokenization [61] yields more interpretable subword units for explanation analysis, while both models achieve comparable performance in preliminary tests.

In our experiments, we fine-tuned a pre-trained BERT model from the *Hugging Face Transformers* library⁴ by training only the classification head on each dataset (AGNews and ACM-4classes), using embeddings from the last hidden layer. Using this representation, we apply k NN with $k = 30$ ⁵ to the test set, identifying the nearest neighbors and computing cosine similarity and Shannon entropy for each instance. The classifier achieves 0.95/0.95/0.95 (Acc/Macro-F1/Wtd-F1) on AGNews and 0.89/0.87/0.89 on ACM. Previous work has demonstrated that, when leveraging very separable embeddings as those produced by BERT, the k NN classifier is as effective as any classifier [3].

We characterize instance locality using two metrics, *similarity* and *entropy*, where similarity is derived from cosine distance, with smaller values indicating higher similarity and larger values indicating lower similarity. Heatmaps visualize the distribution of instances as a function of similarity and entropy, capturing varying degrees of locality homogeneity and highlighting regions with different

⁴ Transformers: <https://huggingface.co/docs/transformers/index>

⁵ This k value optimized k NN classification effectiveness in a validation partition.

neighborhood structures. Two heatmaps are generated for AGNews: one for *correctly classified* (Fig. 1a) and another for *misclassified* instances (Fig. 1b) (based on k NN model classification). ACM heatmaps are provided in (Fig. 2a and 2b). This distinction enables the analysis of how model accuracy and errors affect the explainability of instances when locality is used as an explanatory strategy.

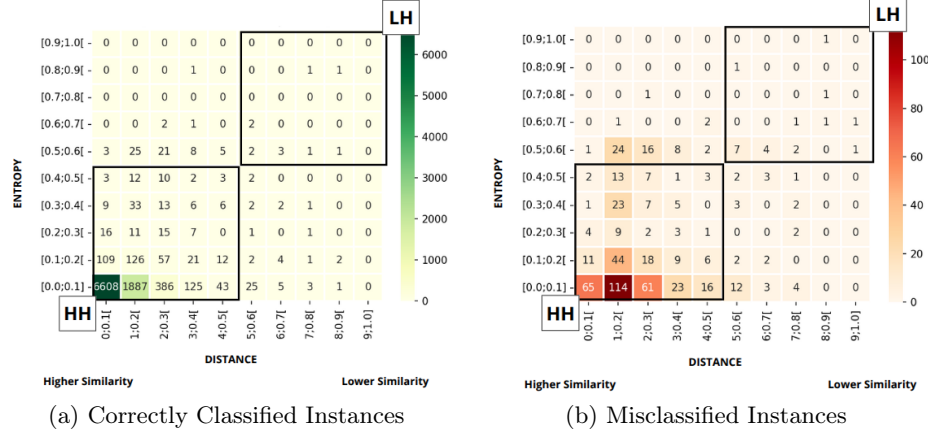


Fig. 1: Heatmaps of Instance Count Distribution (AGNews)

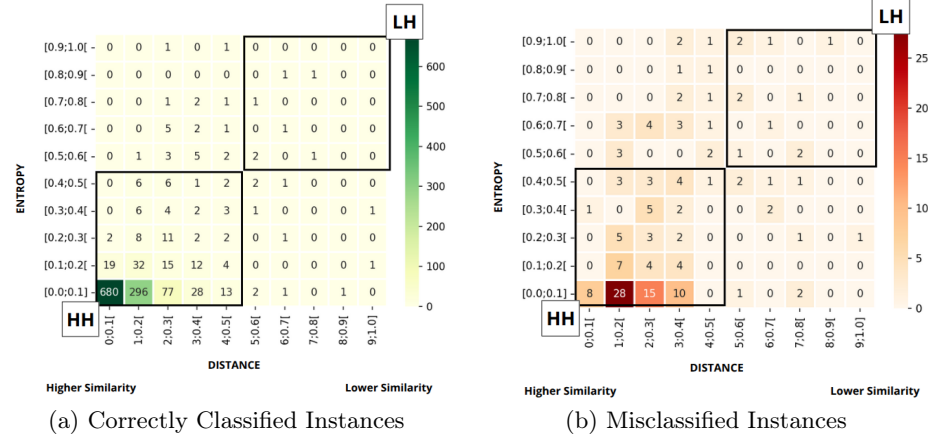


Fig. 2: Heatmaps of instance count distribution (ACM)

Instance Sampling and Explanation Generation

Analyzing the heatmaps (Fig. 1 and 2), we observe that the locality of most instances exhibits low entropy, predominantly concentrated between 0 and 0.1. This can be attributed to the use of transformer-based representations, which enhance class separability, leading to localities with lower entropy values, indicating minimal variation in the class labels of neighboring instances. In each heatmap, two main regions are highlighted: *HH*, denoting *High Homogeneity* (low en-

entropy ≤ 0.5 with low distance ≤ 0.5 , i.e., high similarity), and *LH*, denoting *Low Homogeneity* (high entropy > 0.5 with high distance > 0.5 , i.e., low similarity).

To ensure a diverse selection, we sample 16 representative instances covering both correctly and misclassified cases, sampled from regions of high and low locality homogeneity (HH and LH), and representing all four dataset classes (based on their true labels). For each selected instance, we generate two types of explanations: one based on instance locality, capturing information from neighboring instances, and another derived from the Transformer’s attention weights, reflecting the model’s internal focus during classification; both are represented as *word clouds*. The use of the *same visual metaphor for both types of explanations* avoid any type of **bias** while comparing them.

Very importantly, our adoption of word clouds is a *methodologically grounded and purpose-driven design choice*, not merely an aesthetic one. As discrete, frequency-based visual summaries, This makes them particularly suitable for our goal of relating locality to explanation quality, since they explicitly reveal lexical concentration, topical coherence, and dispersion patterns across regions. Our choice is reinforced by visualization and HCI research showing that carefully curated word clouds; when supported by robust preprocessing, principled weighting, and controlled layouts; can effectively support analytical reasoning and explanation-oriented tasks [26, 27, 55, 60]. Crucially, in our setting, word clouds surface lexical noise, topic fragmentation, and instability in low-homogeneity regions, thereby faithfully exposing the underlying distributional structure rather than imposing artificially coherent narratives, as fluent textual explanations may inadvertently do [10].

Also importantly, our goal is not to claim that word clouds are universally optimal explanation interfaces, but to adopt a familiar and lightweight representation that allows us to control the explanation medium and isolate the effect of semantic locality on users’ perceived explanation quality.

Locality-based word clouds are generated by concatenating neighboring texts into a single document” and extracting frequent terms after preprocessing. After testing different configurations, we set a limit of 25 terms to enhance user comprehension. Preprocessing removes standard *stopwords*, days, months, numbers, URLs, and e-mails, normalizes plurals, and incorporates bigrams. For AGNews, domain-specific *stopwords* are filtered out, including journalistic terms such as *fullquote*”, *quot*”, *reuters*”, and *said*”, as well as agency names like *associated press*”, *(ap)*”, *canadian press*”, and *(cp)*”, which do not contribute to topical content. For ACM, such filtering is unnecessary, as texts are academic abstracts without recurring markers.

Attention-based word clouds, following the same preprocessing, are generated by selecting the 25 terms with the highest attention weights. As an example, Fig. 3b and 3c illustrate word clouds derived from locality and attention weights, respectively, for the news article shown in Fig. 3a, sampled from high-homogeneity region, *correctly classified*, belonging to class *Sports*.

As it can be observed, the locality-based cloud (Fig. 3b) emphasizes sports-related vocabulary, particularly basketball, with terms such as *scored*, *point*, *win*,

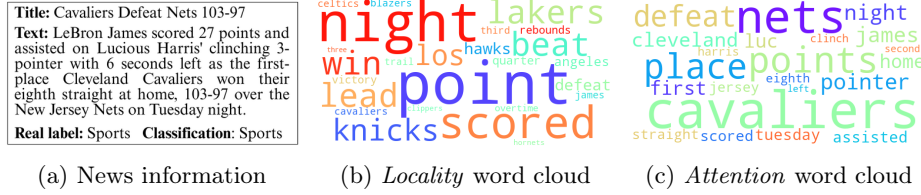


Fig. 3: *Correctly Classified instance with locality- and attention-based explanations.*

lead, and team names like *Lakers* and *Knicks*, strongly aligning with the *Sports* class. In contrast, the attention-based cloud (Fig. 3c), while including words such as *points*, *scored*, and *Cavaliers*, also introduces less discriminative instance-based terms, such as *tuesday*, *place*, and *jersey*, with less team names diversity.

3.3 User Study Design

The user study aimed to evaluate the effectiveness of the generated explanations, represented as word clouds, in clarifying the model’s classification decisions for each dataset, from a user’s perspective. We conducted the same online user study (implemented as a structured survey) for both datasets, following a standardized evaluation procedure described below.

User Study Setup

The user evaluation was conducted through an online survey created with *SurveyMonkey*⁶. The study protocol was approved by the University’s research ethics committee. The survey included an introduction outlining the research objectives, study design, and detailed instructions for the evaluation task. At the end of the introduction, participants were asked to give their informed consent in order to continue and participate in the survey.

Following this introduction, the questionnaire was organized into sequential stages, each corresponding to the evaluation of the two word clouds generated for a single text instance. In each stage, participants were asked to: (i) evaluate *Word Cloud 1*, (ii) evaluate *Word Cloud 2*, and (iii) compare the two word clouds to determine which provided the better explanation for the classification assigned to the text. Each participant could evaluate up to 16 text instances (stages) in each dataset, each accompanied by its corresponding word clouds, and was free to continue or end their participation after any stage. Each stage presented two word clouds; one generated from *locality information* and one from *attention weights*. The presentation order (locality- or attention-based first) was alternated across stages to mitigate positional bias. Each stage also displayed the *title* and *text* of the instance, the model’s predicted class, and a link to a *class description guide*.

Survey Design Considerations

The survey design accounted for two potential risks. First, although the 16 instances were sampled to capture specific characteristics, namely varying locality homogeneity (high/low), classification accuracy (correct/incorrect), and

⁶ <https://www.surveymonkey.com/>

the four dataset classes; the order of presentation could still bias participant responses. To mitigate this, we employed a counterbalancing strategy based on a *Graeco-Latin Square* design [32], controlling for two categorical factors: (1) the instance’s true class (four classes), and (2) its correctness-homogeneity configuration (correct/incorrect $\times$ high/low homogeneity). This design generated a presentation sequence that systematically balanced these factors across stages, thereby reducing potential order effects.

Because participants could leave at any time, later instances might receive fewer evaluations. To mitigate this, for each dataset, we created two survey versions, one with the original Graeco-Latin Square order and one reversed, and evenly distributed them to balance responses.

Survey recruitment and distribution

The survey targeted Computer Science researchers and practitioners with Machine Learning experience, as representatives of the intended user of the proposed explanations. The invitation was sent to potential participants by convenience, and they were encouraged to share it with others who represented the intended audience. For each dataset, the two survey versions were distributed in a balanced manner to ensure a similar number of evaluations per instance.

3.4 Evaluation of Explanation Quality

Evaluating explanation quality in AI systems is inherently challenging, as usefulness depends on users’ goals, prior knowledge, and context. Explanation effectiveness therefore spans multiple dimensions, such as clarity, usefulness, satisfaction, trust, and impact on mental models; which cannot be captured by a single metric. Prior work argues for multi-dimensional, human-centered evaluation frameworks to reflect the subjective and context-dependent nature of explainability [28, 34, 45].

It is worth pointing out that our evaluation does not intend to assess causal faithfulness. The explanations are not intended to reconstruct the classifier’s internal causal mechanisms, but to provide *model-agnostic*, locality-based summaries grounded in the geometric and semantic structure of the embedding space. Consistent with prior distinctions between interpretability and faithfulness [31], our goal is to support human understanding through semantic transparency rather than mechanistic fidelity.

Thus, this work adopts a human-centric evaluation paradigm, focusing on users’ subjective perceptions and judgments, a core class of evaluation metrics in XAI research [28, 34, 45]. Such metrics directly assess whether explanations support understanding and mental-model formation, and are particularly appropriate in early-stage or exploratory studies where application-grounded or task-based evaluations may be impractical or introduce confounding factors [28, 45].

Within this paradigm, we focus on three complementary quality dimensions: *Satisfaction*, *Completeness*, and *Trust*, grounded in prior literature:

- **Satisfaction:** User’s subjective judgment that an explanation is understandable, useful, and adequate for their goals [7, 48–50].

- **Completeness:** Extent to which an explanation includes key factors and sufficient detail to understand a system’s decision-making process [28, 29].
- **Trust:** Calibrated confidence in AI: User’s ability to judge when to rely on model outputs and when to be cautious, based on perceived reliability, predictability, and safety [28, 37, 48, 49].

For each word cloud, participants answered three questions adapted from psychometric instruments reviewed by Hoffman et al. [28]: *Q1: The explanation helped me understand why the model chose this class* (Satisfaction); *Q2: The highlighted words are relevant to the presented class* (Completeness); *Q3: After seeing this explanation, I feel comfortable with the model’s classification* (Trust).

Responses were collected using a five-point Likert scale (*Strongly agree* to *Strongly disagree*), with an additional *Cannot evaluate* option. Such scales offer a simple and effective way to gather structured feedback in human-centered studies, revealing robust trends even with modest sample sizes [66].

We intentionally do not include behavioral or task-based validation (e.g., error detection or counterfactual prediction). Prior work shows that these tasks may introduce confounds related to domain expertise, cognitive load, or task design, and may not align with the intended function of many explanation types [45, 65]. Moreover, subjective judgments and behavioral performance often capture distinct, and sometimes orthogonal, dimensions of explainability [28, 34].

More specifically, [6] shows that performance on simulation-style proxy tasks and self-reported trust do not reliably reflect how explanations influence user decisions in realistic settings. Complementary evidence from [9, 25] indicates that subjective impressions often diverge from objective measures such as simulatability or model comprehension. This mismatch is particularly relevant for explanations primarily intended to support qualitative understanding rather than measurable gains in accuracy [53], as using task performance as the main validation criterion risks misalignment with their intended purpose. Moreover, behavioral-task designs often rely on artificial scenarios that reduce ecological validity and limit generalizability [15]. By focusing on users’ perceptions, our evaluation captures explanations as encountered in more naturalistic interaction settings.

To balance coverage and participant effort, we restricted the evaluation to three dimensions and included an optional open-ended question to capture factors influencing judgments. Such qualitative feedback supports human-centered XAI evaluation, where depth of insight matters more than scale [65, 66]. While task-based evaluations provide complementary evidence, we view them as direction for future work rather than a requirement for our user-centered study [14, 45].

4 Results

4.1 Demographics of Participants

A total of 41 participants completed the survey for AGNews and 33 for ACM. Approximately half were between 18 and 40 years old, with the remainder belonging to older age groups; no minors participated, as eligibility was restricted to adults. Most respondents were researchers or practitioners in Computer Science and related fields, with ML experience ranging from one to more than five

years. On average, participants evaluated eight instances per dataset; those who completed both surveys evaluated eight instances in each dataset.

4.2 Evaluation of Explanations

Each word cloud explanation was evaluated along three criteria (Satisfaction, Completeness, and Trust) using the scale described in Section 3.4. We compare participants’ evaluations of the explanations provided as word clouds in multiple scenarios, each one defined by a pair of instance groups. For each scenario, we begin with a descriptive analysis of positive ratings, defined by combining *Agree* and *Strongly agree* responses, to provide an initial indication of which group received greater support for the use of word clouds under each criterion. To substantiate these observations, we then conducted a statistical comparison of the full response distributions using a chi-square (χ^2) test. This non-parametric test assesses differences between groups when the dependent variable is nominally measured [44]. In our case, the dependent variable corresponds to the distribution of responses across six nominal categories (the five Likert-scale options plus *Cannot evaluate*). For each comparison, we report the associated p-value.

Impact of Locality Homogeneity on Locality-Based Explanations

Based on Fig. 1 (AGNews) and Fig. 2 (ACM), we selected instances from regions of *High Homogeneity* (HH) and *Low Homogeneity* (LH), corresponding to groups 1 and 2 in this comparison scenario, respectively. Each group includes both correctly and misclassified instances. The distribution of user ratings for word clouds, sampled from these two groups (HH e LH), are presented in Fig. 4 for both datasets. For each dataset, AGNews and ACM, Fig. 4 displays bar charts for *Satisfaction*, *Completeness* and *Trust* criteria for each group, namely HH (column (a)) and LH (column (b)). Responses are expressed as percentages over the full scale (Sec. 3.4).

This analysis addresses a data-centric question: *whether the structural region of the representation space from which an instance is sampled, HH versus LH regions, affects user-perceived explanation quality across the three evaluation criteria*. Comparing user ratings for clouds from *High* (HH) and *Low-Homogeneity* (LH) regions, we observe that HH clouds consistently receive a higher proportion of positive ratings across all evaluation criteria in both datasets. For AGNews, the positive rating rates for HH vs. LH regions are: *Satisfaction* - 76.6% vs. 36.4%, *Completeness* - 78.3% vs. 40.6%, and *Trust* - 72.3% vs. 35.2%. For ACM, the values are: *Satisfaction* - 86.2% vs. 73.6%, *Completeness* - 84.5% vs. 69.1%, and *Trust* - 83.3% vs. 66.4%.

These differences are statistically significant in both datasets. χ^2 tests of independence on the full scale distributions provide strong evidence of differences between HH and LH ratings across all three criteria ($p < 0.001$) for both AGNews and ACM. Notably, when comparing positive ratings from HH and LH, the gaps are consistently larger in AGNews.

These findings indicate that explanations drawn from HH regions are perceived as more effective and easier to understand. Moreover, they strengthen the

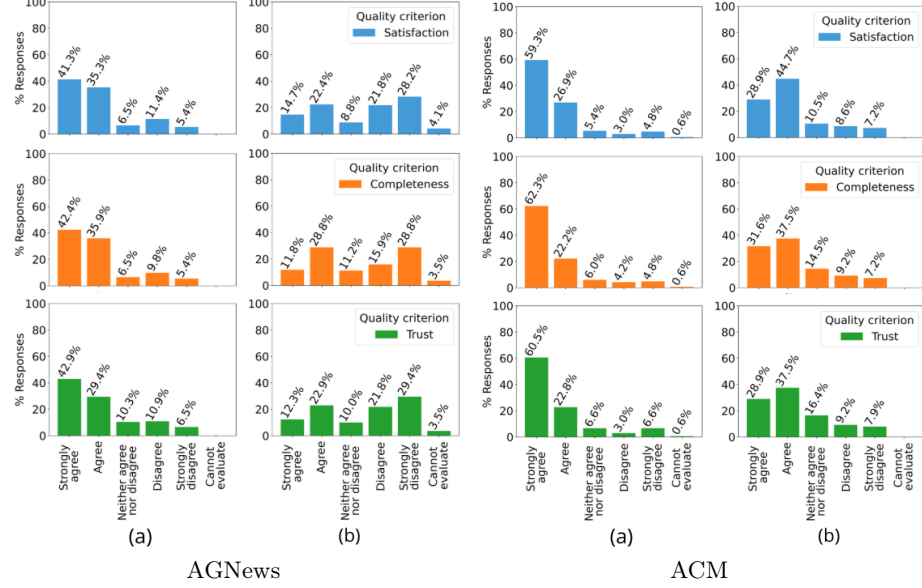


Fig. 4: Distributions of user ratings for locality-based word clouds - (a) *High-Homogeneity*(HH) and (b) *Low-Homogeneity*(LH) Regions

generalizability of our results and support our hypothesis that more homogeneous localities tend to yield more interpretable explanations.

(i) *Correctly Classified Instances*

In this scenario, we restrict the analysis to *correctly classified* instances (Fig. 5), and again compare instances from the HH and LH regions. We observe the same pattern: locality-based word clouds from *HH* regions receive a higher proportion of positive ratings than those from *LH* regions across all three criteria. χ^2 tests on the full scale distributions confirm significant *HH-LH* differences for all criteria in AGNews ($p < 0.01$) and ACM ($p < 0.001$). Thus, statistical significance is stronger in ACM, remaining consistent across both datasets.

The same pattern observed across news and academic texts suggests that the effect may generalize beyond a single domain. Its stronger impact in ACM indicates that homogeneous locality may be especially valuable in specialized or complex settings where model behavior is harder to interpret.

(ii) *Misclassified Instances*

Restricting the analysis to *misclassified* instances (ratings shown in Fig. 6), we observe a similar but weaker pattern. In this scenario, when comparing positive ratings between HH and LH, HH-LH gaps remain positive across all three criteria in AGNews, although smaller in magnitude, with χ^2 tests indicating significant *HH-LH* differences for all three ($p < 0.001$). In ACM, positive ratings were slightly higher for HH than for LH; however, the χ^2 tests revealed no reliable difference for *Satisfaction* ($p > 0.05$), while showing significant differences for *Completeness* and *Trust* ($p < 0.05$). These results suggest that the positive im-

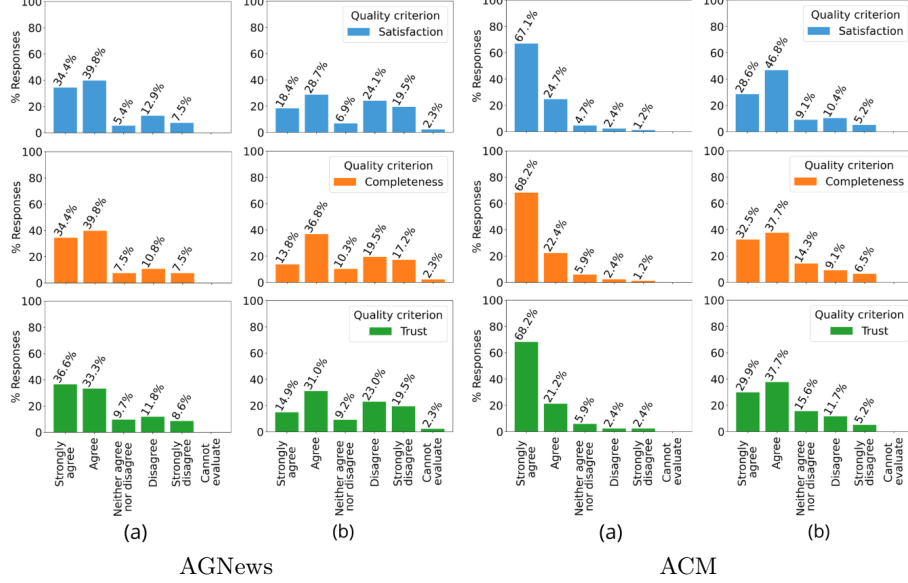


Fig. 5: User Ratings Distribution for Locality WordClouds - (a) *High-Homogeneity*(HH) and (b) *Low-Homogeneity*(LH) - **Correctly Classified**

impact of *HH* regions on explanation quality persists even in misclassified instances, indicating that the explanations effectively convey why the system produced a given classification, although the effect is attenuated compared to correctly classified instances. This is understandable as errors may be harder to explain.

Impact of Prediction Correctness on Locality-Based Explanations

This analysis addresses a model-centric question: *whether the correctness of the model's prediction, correctly classified versus misclassified instances, affects user-perceived explanation quality across the three evaluation criteria*. We examine whether explanations for correctly classified instances are systematically evaluated more favorably than those for misclassified ones, or whether no such dependency is observed. To investigate this issue, we focus on locality-based explanations and partition them into two groups: those generated for *correctly classified* and *misclassified* instances (groups 1 and 2, respectively). In this complementary analysis, user evaluations in each group are compared for each criterion using chi-square tests, and corresponding p-values are reported.

In the ACM dataset, chi-square tests reveal that the perceived explanation quality is independent of prediction correctness across all criteria ($p > 0.05$ for all criteria). For AGNews, a significant association emerges only for *Completeness* ($p \leq 0.05$), while for *Satisfaction* and *Trust* remain non-significant ($p > 0.05$).

When the analysis is restricted to explanations of instances drawn exclusively from high-homogeneity (HH) regions (comparing correctly and misclassified instances from HH regions), user evaluations are independent of prediction correctness across all quality criteria (*Satisfaction*, *Completeness*, and *Trust*) in both the AGNews and ACM datasets, with all comparisons yielding non-significant

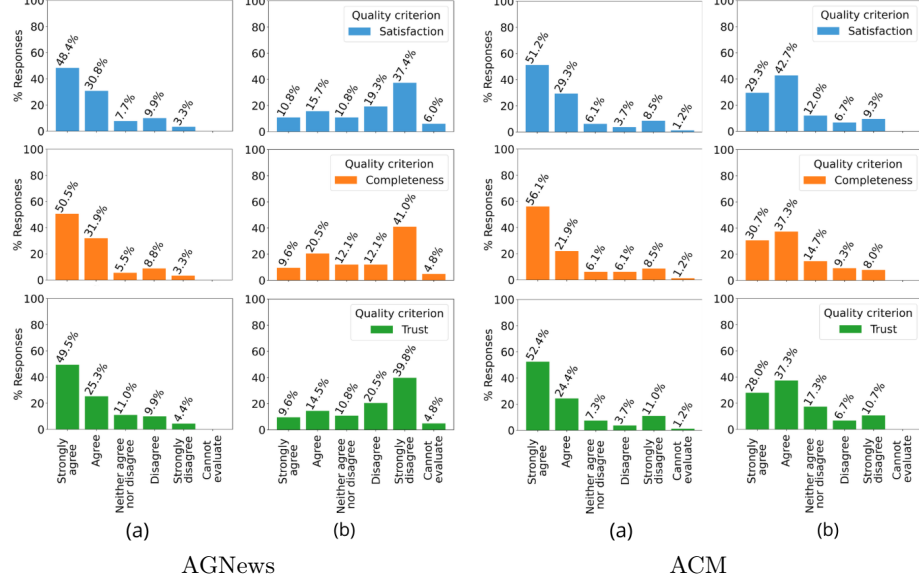


Fig. 6: Distribution of User Ratings for Locality Wordclouds - (a) *High-Homogeneity*(HH) and (b) *Low-Homogeneity*(LH) Regions - **Misclassified**

results ($p > 0.05$). In contrast, when the analysis is restricted to low-homogeneity (LH) regions, user evaluations in the ACM dataset remain independent of prediction correctness across all criteria. For AGNews, however, prediction correctness is significantly associated with all evaluated criteria in LH regions ($p \leq 0.05$ for Satisfaction and Trust, and $p \leq 0.01$ for *Completeness*).

The results indicate that the relationship between prediction correctness and user-perceived explanation quality is indirect and mediated by dataset structure and locality homogeneity, reinforcing our hypothesis. In semantically homogeneous (HH) localities, explanations tend to be evaluated favorably regardless of prediction correctness, as surrounding neighborhoods provide a coherent semantic context that supports interpretation. In contrast, in low-homogeneity localities, particularly in AGNews with its broad topical categories and diffuse boundaries, misclassifications negatively affect perceived completeness, satisfaction, and trust, as explanations reflect greater lexical and thematic ambiguity. For ACM, however, specialized technical vocabulary and more sharply defined semantic structures mitigate this effect even in less homogeneous regions, rendering explanation quality largely independent of prediction correctness.

Taken together, these results indicate that user-perceived explanation quality is not systematically driven by prediction correctness. Explanations associated with correctly classified instances are not, in general, evaluated more favorably than those associated with misclassified ones. Instead, the influence of correctness on user judgments emerges only in LH regions of the AGNews dataset, presenting structurally ambiguous settings.

Comparison of Locality-Based and Attention-Based Explanations

In this additional scenario, we compare two groups of explanations: locality and attention-based word clouds (groups 1 and 2, respectively), as a complement to the main study. It is worth noticing that both explanations may convey complementary information: locality-based explanations capture regional characteristics of the feature space surrounding an instance, whereas attention-based explanations focus on the instance’s specific, fine-grained features.

In AGNews, the proportions of positive ratings were very similar between locality and attention-based word clouds, including correctly and incorrectly classified instances within each group (ratings shown in Fig. 7). χ^2 tests indicated a significant difference only for *Satisfaction* ($p \leq 0.05$), with no reliable differences for other criteria; for *Satisfaction*, positive ratings were slightly higher for attention clouds. Nonetheless, when participants made a forced-choice selection (Fig. 8), locality explanations were chosen by 39.6% of users versus 33.4% for attention-based ones, with 22.3% judging them equivalent. This indicates that, despite modest criterion-level differences, locality seems to offer a small advantage when users must pick the explanation that best supports the model’s prediction.

In ACM, locality-based explanations yielded higher proportions of positive ratings across *Satisfaction*, *Completeness*, and *Trust* when compared with attention ones. χ^2 tests on the full scale distributions revealed significant differences for all three criteria, strongest for *Satisfaction* ($p \leq 0.01$) and still significant for *Completeness* and *Trust* ($p \leq 0.05$). When asked to choose one explanation (Fig. 8), 45.9% preferred the locality-based word cloud, 22% the attention-based one, and 26.8% found them equivalent, confirming locality’s advantage on ACM.

Satisfaction was the most discriminative criterion: in AGNews, the two methods received similar ratings, with only a marginal tendency in favor of locality; in ACM, locality explanations scored significantly higher. This pattern suggests that locality’s benefit may be modest in simpler domains but becomes more evident as domain complexity increases, yielding gains for more technical texts. This difference reflects the semantic structure of the datasets: AGNews contains broader topics with overlapping vocabulary, whereas ACM exhibits more specialized terminology, making locality-based aggregation more informative.

ACM is composed of fine-grained, technical subfields (e.g., *Theory of Computation*, *Hardware*, *Information Systems*) where documents rely on specialized terminology that is tightly clustered in the embedding space. In this setting, *Locality* aggregates domain-specific vocabulary across semantically similar papers, revealing stable, higher-level conceptual themes. *Attention*, being instance-specific, may highlight isolated technical details that matter for classification but do not generalize across the neighborhood, making explanations feel more fragmented. Because ACM exhibits clearer semantic clusters, locality exposes more coherent domain patterns than attention-based explanations.

4.3 Discussion and Implications

Our results indicate an empirical association between locality structure and perceived explanation quality, rather than causal proof of improved explainability.

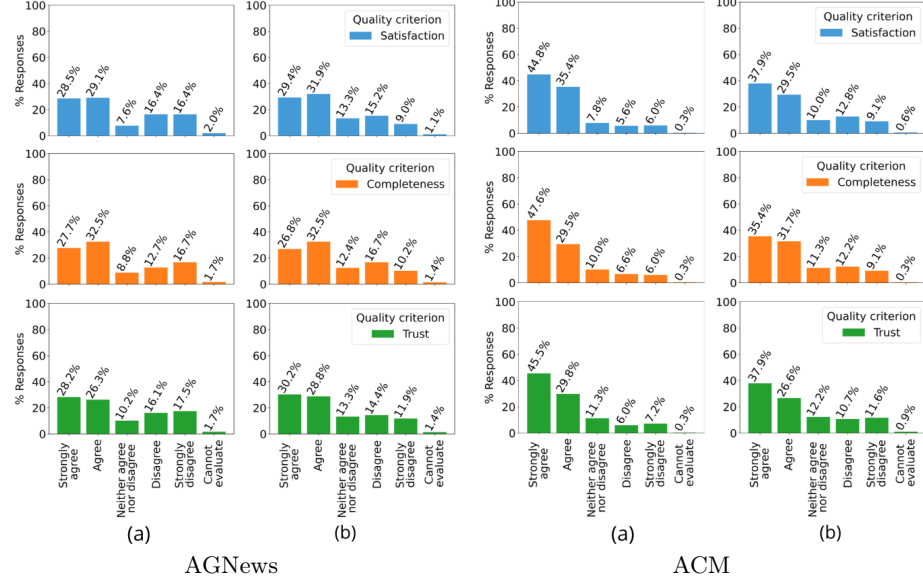


Fig. 7: Distribution of User Ratings for Locality and Attention word clouds

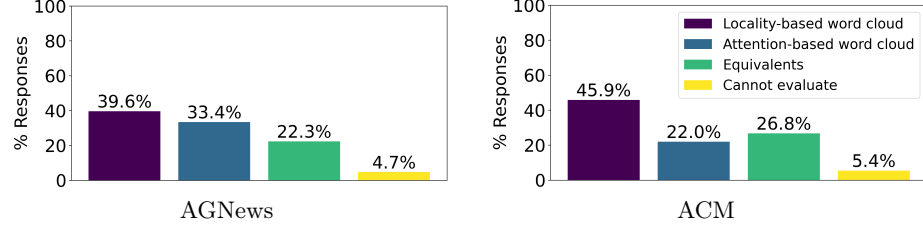


Fig. 8: Distribution of user preferences: locality vs. attention word clouds

Differences in user-perceived explanation quality across ACM and AGNews can be attributed to their distinct semantic granularity and neighborhood structure:

(i) *Stronger Homogeneity Effects in ACM*: In ACM, neighborhood homogeneity has a pronounced impact: high-homogeneity instances receive significantly higher ratings across *Satisfaction*, *Completeness*, and *Trust* (all $p < 0.001$). The dataset’s fine-grained technical subfields form tight semantic clusters, making coherent neighborhoods crucial for producing interpretable and trustworthy explanations. In contrast, low-homogeneity regions combine disparate technical vocabularies, yielding less coherent explanatory signals.

(ii) *Weaker but Consistent Effects in AGNews*: AGNews exhibits the same trend, but with smaller effect sizes. News articles are shorter and stylistically uniform, and even low-homogeneity neighborhoods often share familiar vocabulary. This lexical overlap reduces the perceptual contrast between explanations derived from homogeneous and heterogeneous regions, attenuating locality effects.

(iii) *Domain Sensitivity for Misclassified Instances*: Locality effects are more nuanced for misclassified instances. In AGNews, high-homogeneity instances con-

sistently outperform low-homogeneity ones across all criteria. In ACM, however, homogeneity improves *Completeness* and *Trust* but not *Satisfaction*, suggesting that misclassified technical content demand higher cognitive load, even under coherent neighborhood context.

(iv) *Locality versus Attention across Domains*: Comparisons with attention-based explanations reinforce this interpretation. In AGNews, locality-based and attention-based explanations perform similarly, capturing overlapping surface-level cues. In ACM, locality-based explanations consistently outperform attention-based ones, indicating that aggregating evidence from semantically coherent neighborhoods is more informative and trustworthy than instance-level attention in specialized domains.

Overall, these results show that user-perceived explanation quality is shaped by domain complexity and neighborhood coherence. Locality-based explanations are particularly advantageous in datasets with fine-grained, tightly clustered semantics, while the influence of prediction correctness is conditional and emerges primarily under low-homogeneity conditions in domains such as AGNews.

5 Conclusion

We examined the role of locality in enhancing ATC explainability by characterizing neighborhoods through *similarity* and *entropy*. Results show that homogeneous localities (high similarity, low entropy) produce explanations perceived as clearer and more trustworthy, providing empirical support for the assumed benefits of locality.

Beyond this general effect, neighborhood homogeneity predicts explanation quality in systematic, dataset-dependent ways, with stronger effects in ACM and smaller but significant effects in AGNews. These patterns extend to misclassified instances, revealing that heterogeneous neighborhoods do not uniformly degrade explanations, and that misclassified instances in highly homogeneous regions often yield locally coherent explanations. This suggests that some errors stem from annotation noise or boundary ambiguity rather than model failure alone.

These findings yield several practical implications for XAI practitioners:

- i. *Locality-informed instance selection*: Locality metrics help identify instances likely to yield informative explanations, supporting introspection, communication, and data auditing.
- ii. *Locality-aware evaluation*: Stratifying evaluations by neighborhood homogeneity enables more transparent and reliable assessment.
- iii. *Anticipating explanation failures*: Locality properties enable early detection of regions expected to exhibit reduced explanation quality, supporting adaptive strategies.
- iv. *Dataset-level interpretability analysis*: Locality provides a lens to assess dataset-level interpretability, highlighting how noisy or fragmented neighborhoods impose independent limits.

By grounding explainability in the semantic structure of the data manifold rather than model-specific mechanisms, our framework yields insights that gen-

eralize across classifiers and explanation techniques. This model-agnostic perspective supports transfer across architectures and tasks, linking user-centric explanation quality to measurable distributional properties. We position locality as a foundational concept for the design, evaluation, and deployment of explainable ATC systems. From the perspective of established XAI properties [14,28,31,45,47,65], our data-centric approach contributes in complementary ways. Locality homogeneity supports *Stability*, while neighborhood aggregation enhances *Expressiveness*. This design improves clarity but entails a trade-off with *Faithfulness*. Word clouds provide *Translucency*, and the representation-based mechanism supports *Portability*. Finally, neighborhood aggregation promotes *Plausibility* by emphasizing semantically coherent terms, without guaranteeing direct causal correspondence.

Future work includes extending the analysis to additional datasets and domains, studying class-specific effects, incorporating richer qualitative feedback, and exploring alternative formats, such as LLM-generated summaries, to complement locality-based visual explanations, as well as investigating connections between locality-based word clouds and qualitative analysis approaches such as grounded theory and topic modeling, and the sensitivity of locality characterization to alternative embedding models, distance metrics, and neighborhood sizes.

Limitations

While our study offers novel insights, it presents some natural scope constraints. First, the methodology was evaluated on two text classification datasets (ACM and AGNews), which, although more informative than a single-domain setup, still limit generalization. Extending the analysis to additional datasets and domains is a promising direction; however, large-scale qualitative user studies across multiple benchmarks entail substantial cost, logistical complexity, and participant recruitment challenges.

Second, we focused on word clouds as our explanation medium. This choice was methodologically grounded, as word clouds provide transparent, auditable, and easily comparable summaries of salient terms, enabling systematic analysis of how locality shapes explanatory content. They proved effective for exposing lexical concentration, topical coherence, and fragmentation patterns across regions, although they do not capture the full contextual structure. For new instances, nearest-neighbor search may add computational cost, but only the incoming instance embedding must be generated. Future work may compare them with alternative representations, including LLM-generated summaries grounded in neighborhood information; however, such comparisons should be conducted carefully, given the risk of interpretative bias and hallucination in free-form narratives.

Finally, our evaluation relied on user-perceived explanation quality collected via a survey and did not assess mechanistic or causal faithfulness with respect to the classifier’s internal decision process. Increasing participant diversity and scale, alongside richer qualitative analyses, may deepen understanding of how

different users interpret locality-based explanations and further strengthen the robustness and applicability of our framework.

6 Ethical Considerations

This study adheres to established ethical standards for research involving human participants and was reviewed and approved by the Research Ethics Committee of the Federal University of Minas Gerais⁷. Participation was voluntary, informed consent was obtained, and participants could withdraw at any time. No personally identifiable information was collected, responses were anonymized and stored for research purposes. Given the study non-sensitive nature, risks were minimal, and privacy and transparency were ensured in accordance with Brazilian regulations and international standards for responsible AI research and evaluation.

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⁷ CAAE ID: 44032821.0.0000.5149

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A Class Description Guide

AGNews Dataset. Contains short news texts. The model identifies the main topic of each document among four categories:

- **Sports:** News on sports, athletes, competitions, championships, and related events.
- **Science/Technology:** News on technology, innovations, gadgets, research, and scientific developments.
- **World:** News on international events, politics, and global or country-related affairs.
- **Business:** News on economics, financing, companies, and business-related topics.

ACM Dataset. Comprises Computer Science research texts. This survey considers four of the eleven original categories:

- **Information Systems:** Design and application of information systems, including decision support, web systems, e-commerce, GIS, databases, interfaces, and DLs.
- **Hardware:** Physical components and architectures of computing systems, including circuits, microprocessors, devices, and performance optimization.
- **Software Engineering:** Software development, including languages, modeling, testing, validation, reuse, configuration management, and quality.
- **Theory of Computation:** Mathematical foundations of computing, including algorithms, computability, complexity, automata, logic, and cryptography.